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14.1 INTRODUCTION

Before the commencement of any project, it is necessary to work out the probable cost of construction. This is known as the estimated cost. Accuracy of estimation depends upon the accuracy of drawing, specifications and assumptions.

The norms of estimation considered in this chapter are based on practical working experience with projects completed as on February 1, 2001. The constants, rates, methods etc. may vary from place to place, organization to organization and project to project.

14.2 USES OF ESTIMATION

- Estimation gives a fairly accurate idea of the project cost before the execution of the work can begin.
- The cost is worked out after considering the drawings and the specifications.
- The material quantities and material schedules are worked out.
- Arrangements for the labour force can be made after considering the specified work period.
- Bar charts and cash flow combinations can be prepared for smooth progress of the project.
- Once the estimation is ready, funds can be made available.
- It helps to source the plants and machinery required at the project site.

14.3 TYPES OF ESTIMATES

The many types of estimates can be grouped into two main categories -

(a) Approximate estimates
(b) Detailed estimates

An approximate or rough estimate is prepared in a short time to get a rough idea of the cost. This estimate may be treated as the preliminary 'Quick Method' estimate.

In detailed estimates, cost is worked out considering all the quantities in detail along with other factors.

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Therefore, this estimate is reliable and accurate.

14.4 PRELIMINARY "QUICK METHOD" ESTIMATION

Generally 'Quick Method' estimates are based on the practical knowledge and the costs of similar completed works. The materials required for the project can be obtained using the material consumption constants based on the earlier projects.
e.g.
Required material quantity = Built-up area x respective material consumption constant

Preliminary 'Quick Method' estimation is required for preliminary studies of the various aspects of project.

- It gives a rough idea of the probable expenditure.
- It helps in securing the administrative approval.
- It helps in deciding the selling value/rate.

(1) DATA REQUIRED FOR PRELIMINARY ESTIMATION

- A complete set of working drawings.
- Specifications of the various items.
- Consideration of other development activities like approach roads, compound wall, water tank, connection to main drainage line etc.
- 'Built-up area' of the project.
- Prevailing rates of the material and labour.

(2) COMMON PRACTICE FOR BUILT-UP AREA CALCULATIONS

Preliminary estimate is calculated on the basis of built-up area. Approximate built-up area for estimation is calculated as follows -

- Parking/stilt areas are taken as 50%.
- Ground and all other floors are taken as 100% and no elevation features are considered.
- Over head water tank base area is taken as 50%.
- Balconies, staircase, lift areas are taken as 100%.

(3) VARIOUS HEADS TO BE CONSIDERED FOR APPROXIMATE 'QUICK METHOD' ESTIMATION

Cost of the project depends on various major and minor heads of expenditure. All these heads can be broadly summarized in two main heads as follows-

(A) DIRECT COST
(B) INDIRECT COST

The points to be considered under these heads are,

(A) DIRECT COST

Direct cost may be defined as the actual expenses incurred on the construction of project i.e. the cost of land (known at initial stage), cost of material and labour involved in the work.

- **LAND COST**

Land cost varies considerably from one location to another, even in the same city.

There is no thumb rule for the land cost. Hence, land cost should be added at actuals, considering the expenses under the following heads -

- Legal charges
- Brokerage
- Tenant shifting (if applicable)

- **MATERIAL EXPENSES**

All materials required for the work fall under this category.

Grey cement, white cement, steel, sand, metal, rubble, murum, stone dust, grit, bricks, blocks, sanla, lime, door frames, door shutter, door fittings, windows, grills, tiles, stone materials, plumbing materials and other miscellaneous materials.

- **LABOUR EXPENSES**

R.C.C. work (including plinth work), Masonry and plaster work, Water proofing work, Plumbing work, Tile fixing work, Carpentry work, Electrical work, (including material). Painting work, (including material). Departmental labourers etc.

(B) INDIRECT EXPENSES

Indirect cost may be defined as the correlated expenses incurred for the construction of the building, other than material, labour and land cost. The following points should be considered for various heads of indirect expenses.

■ **SANCTION EXPENSES**

- Development Charges (D.C.) + 7/12 extract expenses
- N.A. Expenses
- U.L.C. Expenses
- Municipal taxes
- Hardship charges

■ **PLOT DEVELOPMENT**

- Plot cleaning and levelling
- Excavation, rock cutting etc.
- Murum and debris filling
- Approach road to plot
- Hoarding boards, tree guards
- *Nalla*/other developments
- Tree cutting expenses

■ **SITE DEVELOPMENT**

- Initial site development
- Compound wall and fencing
- Internal road/Approach road
- U.G.W.T. pump house and bore well
- Swimming pool, health club etc.
- Garden and play ground equipments
- Water lines, drainage line and septic tanks
- Electrical cabling, transformer plinth, and L.T. room
- Street lights

■ **CONSULTANCY EXPENSES**

- Architect
- Structural consultant
- Plumbing and sanitation consultant
- Electrical consultant
- Road consultant
- Landscape designer
- Model/perspective maker
- Other consultants

■ **ADMINISTRATIVE EXPENSES**

- Advertising
- Printing and stationery
- Office expenses
- Office staff remuneration
- Brokerage, donations etc.
- Site office expenses

■ **SUPERVISION EXPENSES**

- Expenses of engineering department
- Site staff remuneration
- Security expenses
- Resident architect expenses etc.

■ **SITE RUNNING EXPENSES**

- Temporary sheds
- Scaffolding and other materials
- Machinery running/maintenance expenses
- Telephone and electricity expenses
- Water and temporary lights expenses
- Staff welfare, insurance and medical expenses
- Other miscellaneous cash purchases
- Day to day site running expenses

■ **MISCELLANEOUS EXPENSES**

- Anti-termite/other treatments
- Any other expenses which do not come under the other heads
- Additional costs due to changes in specification

(4) STANDARD MATERIAL CONSUMPTION CONSTANTS ON BUILT-UP AREA BASIS

(Refer Table No. 14.2 & 14.3)

All the "material consumption constants" given in table no. 14.2 are for reference and act as a guideline, to prepare estimation by the "quick method". Material specifications and consumptions are based on different completed projects in Pune. Materials required for development work have not been considered as consumption constants.

Specifications for the major items are given in Table No. 14.1, to work out the approximate material consumption constants.

MAJOR SPECIFICATIONS OF THE PROJECT

TABLE NO. 14.1

Sr. No.	DESCRIPTION OF ITEM		SPECIFICATIONS FOR MEDIUM CLASS BUILDING
01	Structure		R.C.C. framed structure (ground + 2 floors and parking + 3 floors) concrete M15 grade.
02 (a)	Masonry **Block size** 30cm x 15cm x 20cm (12" x 6" x 8")		External wall 15cm (6") thick block work in C.M. 1 : 6
	30cm x 10cm x 20cm (12" x 4" x 8")		Internal wall 10cm (4") thick block work in C.M. 1 : 4 **or**
(b)	**Brick size** 23cm x 15cm x 9cm (9" x 6" x $3\frac{1}{2}$")		External wall 15cm (6") thick brick work in C.M. 1: 5
	23cm x 10cm x 7cm (9" x 4" x $2\frac{3}{4}$")		Internal wall 10cm (4") thick brick work in C.M. 1 : 4 **or**
			External wall 23cm (9") thick brick work in C.M. 1 : 6
			Internal wall 10cm (4") thick brick work in C.M. 1 : 4
03	Plastering		Sand faced double coat plaster for external walls in C.M. 1 : 5
			Neeru finished plaster for all internal walls in C.M. 1 : 6
			Sand faced single coat plaster for staircase and balconies.
04	Flooring & tiling	Rooms, balconies & passage	Grey marble mosaic tile flooring 25cm x 25cm Grey marble mosaic matching skirting 25cm x 12.5cm
		Bath room	Polished Tandoor stone flooring (machine cut)
		W.C.	Glazed (white) tile flooring 15cm x 15cm
		Parking	Chequered or shahabad flooring
		Staircase	Tappa and landing in Tandoor flooring.
05	Dado for	W.C.	White glazed tiles 15cm x 15cm (6" x 6") up to 60cm (2'0") height
		Bath room	Coloured glazed tiles 15cm x 15cm (6" x 6") up to 1.05m (3'6") height
		Kitchen otta	White glazed tile 15cm x 15cm up to 60cm (2'0") height.
06	Door frames		Door frames in teak wood of section 10cm x 6.5cm (4" x $2\frac{1}{2}$").
07	Door shutters		Commercial type flush door shutter 30mm thick. Main door one side teak veneered.
08	Door fittings		Iron oxidised or Aluminium.
09	Windows		M.S. windows with obscured (Bajari) glass.
10	Plumbing		G.I./P.V.C./C.I. pipes and C.P. fittings, concealed plumbing in bathroom/W.C.
11	Electrical works		Casing-capping wiring..
12	Painting		Dry distemper onwalls, oil paint to door frames, shutters and M.S. windows white wash to ceilings and cement paint to external walls and staircase.

☛ STANDARD MATERIAL CONSUMPTION CONSTANT ON BUILT-UP AREA BASIS

TABLE NO. 14.2 **FOR EXTERNAL WALL 15 cm (6") THICK BLOCK WORK AND INTERNAL WALL 10 cm (4") THICK BLOCK WORK**

SR. NO.	MATERIAL	ITEM OF WORK	QUANTITY PER SQM	QUANTITY PER SFT
01	Cement	For R.C.C. work	1.94 bags	0.18 bag
		For Masonry work	0.43 bag	0.04 bag
		For Plastering work	1.18 bags	0.11 bag
		For Other works	0.32 bag	0.03 bag
		Total	**3.87 bags**	**0.36 bag**
02	Steel	For R. C. C. Work	24.25 Kg	2.25 Kg
03	Sand	For R.C.C. work	0.21 cum	0.70 cft
		For Masonry work	0.05 cum	0.16 cft
		For Internal plaster	0.23 cum	0.75 cft
		For External plaster	0.15 cum	0.50 cft
		For Tiling	0.04 cum	0.13 cft
		For Water proofing	0.030 cum	0.09 cft
		For Others	0.01 cum	0.02 cft
		Total	**0.72 cum**	**2.35 cft**
04	Metal	For R.C.C. Work	0.27 cum	0.90 cft
05	Rubble	For plinth soling	0.06 cum	0.20 cft
06	Murum	For plinth filling	0.23 cum	0.75 cft
07	Bricks (23cm x 10 cm x 7cm) (9" x 4" x 2 $\frac{3}{4}$")	Only for plinth	43 Nos	4 Nos
08	Bricks (Gattu / Thokla) (23cm x 15cm x 9cm) (9" x 6" x 3 $\frac{1}{2}$")	For External wall 15cm (6") thick	43 Nos	4 Nos
		For Internal wall 15cm (6") thick	32 Nos	3 Nos
		Total	**75 Nos ***	**7 Nos***
09	Bricks (23cm x 15cm x 9cm) (9" x 6" x 3$\frac{1}{2}$")	For External wall 15cm (6") thick	43 Nos	4 Nos
	(23m x 10cm x 7cm) (9" x 4" x 2$\frac{3}{4}$")	For Internal wall 10m (4") thick	43 Nos	4 Nos
		Total	**86 Nos ***	**8 Nos ***
10	Blocks (30cm x 15cm x 20cm) (12" x 6" x 8")	For For External wall 15cm (6") thick	14.50 Nos	1.35 Nos
	(30cm x 10cm x 20cm (12" x 4" x 8")	For Internal wall 10 cm (4") thick	13.0 Nos	1.20 Nos
		Total	**27.5 Nos**	**2.55 Nos**
11	Add as extra for cement	If concrete blocks are manufactured at site For block	0.65 bag	0.06 bag
	sand	For block	0.12 cum	0.40 cft
	metal	For block	0.12 cum	0.40 cft
	Grit and stone dust	For block	0.17 cum	0.55 cft
12	Sanla	For Internal plastering	0.27 bag	0.025 bag
13	Lime	For flooring	0.16 bag	0.015 bag
14	M. S. Windows	————	0.14 sqm	0.14 sft
15	Glass for M. S. Windows	————	0.14 sqm	0.14 sft
16	Mosaic tiles	For Rooms & Balcony flooring	0.75 sqm	0.75 sft

Note : * Indicates the constant for various combinations for brick masonry work

SR. NO.	MATERIAL	ITEM OF WORK	QUANTITY PER SQM	QUANTITY PER SFT
17	Mosaic skirtings	For Rooms skirting	0.82 rm	0.25 rft
18	Bath / Toilet Flooring		0.05 sqm	0.05 sft
19	Dado	Bath / W.C. dado including W. C. Flooring		
	a) White glazed tiles		0.09 sqm	0.09 sft
	b) Colour gazed tiles		0.11 sqm	0.11 sft
		Total	**0.20 sqm**	**0.20 sft**
20	Kitchen platform		0.04 rm	0.012 rft
21	Chequered tiles or rough shahabad	parking area in building	0.15 sqm	0.15 sft
22	Staircase Stone Flooring	For Risers	0.014 sqm	0.014 sft
		For Treads	0.028 sqm	0.028 sft
		For Loadings	0.022 sqm	0.022 sft
		Total	**0.064 sqm**	**0.064 sft**
23	Brick bats	For waterproofing	0.03 cum	0.10 cft
24	Door frames	—	0.09 Nos	0.0085 Nos
25	Door Shutters	—	0.14 sqm	0.14 sft
26	Concrete grills	For Staircase	0.03 sqm	0.3 sft

☛ **STANDARD MATERIAL CONSUMPTION CONSTANT ON BUILT-UP AREA BASIS**

FOR EXTERNAL WALL 23 cm (9") THICK BRICK WORK AND INTERNAL WALL 10 cm (4") THICK BRICK WORK

NOTE : All other material remains same as per table no. 14.2 except as mentioned below.

TABLE NO. 14.3

SR. NO.	MATERIAL	ITEM OF WORK	QUANTITY PER SQM	QUANTITY PER SFT
01	Cement	For R.C.C. work	2.04 bags	0.19 bag
		For Masonry work	0.86 bag	0.08 bag
		For Plastering work	1.61 bags	0.15 bag
		For Others	0.33 bag	0.03 bag
		Total	**4.84 bags**	**0.45 bag**
02	Sand	For R.C.C. work	0.22 cum	0.75 cft
		For Masonry work	0.11 cum	0.36 cft
		For Internal Plaster	0.30 cum	1.00 cft
		For External Plaster	0.2 cum	0.65 cft
		For Tiling	0.04 cum	0.13 cft
		For Waterproofing	0.03 cum	0.09 cft
		For Others	0.01 cum	0.02 cft
		Total	**0.91 cum**	**3.00 cft**
03	Bricks (23cmx10cmx7cm) (9" x 4" x 2 $\frac{3}{4}$")	For Plinth work	43.00 Nos	4.0 Nos
		For External wall	86.00 Nos	8.0 Nos
		For Internal wall	43.00 Nos	4.0 Nos
		Total	**172.00 Nos**	**16.0 Nos**

14.5 _ ESTIMATION FOR COMPLETE PROJECT AT A GLANCE (SPECIMEN)

- Preliminary estimation of a total project can be done in the initial stages without any detailed drawings.
- Built-up area should be calculated from the preliminary drawings.
- Specifications of the project work can normally be decided at initial stage of work.
- Total estimation of the project can be done from the material and labour constants per unit of the built-up area.

FOLLOWING MAJOR HEADS ARE CONSIDERED FOR PRELIMINARY ESTIMATION AT A GLANCE

Project Reference : Typical medium class project at Pune. Structure G+2 and parking +3

Drawings : *(Refer Figure No. 14.1)*

Specifications : *(Refer Table No. 14.1)*

Built-up area calculation :

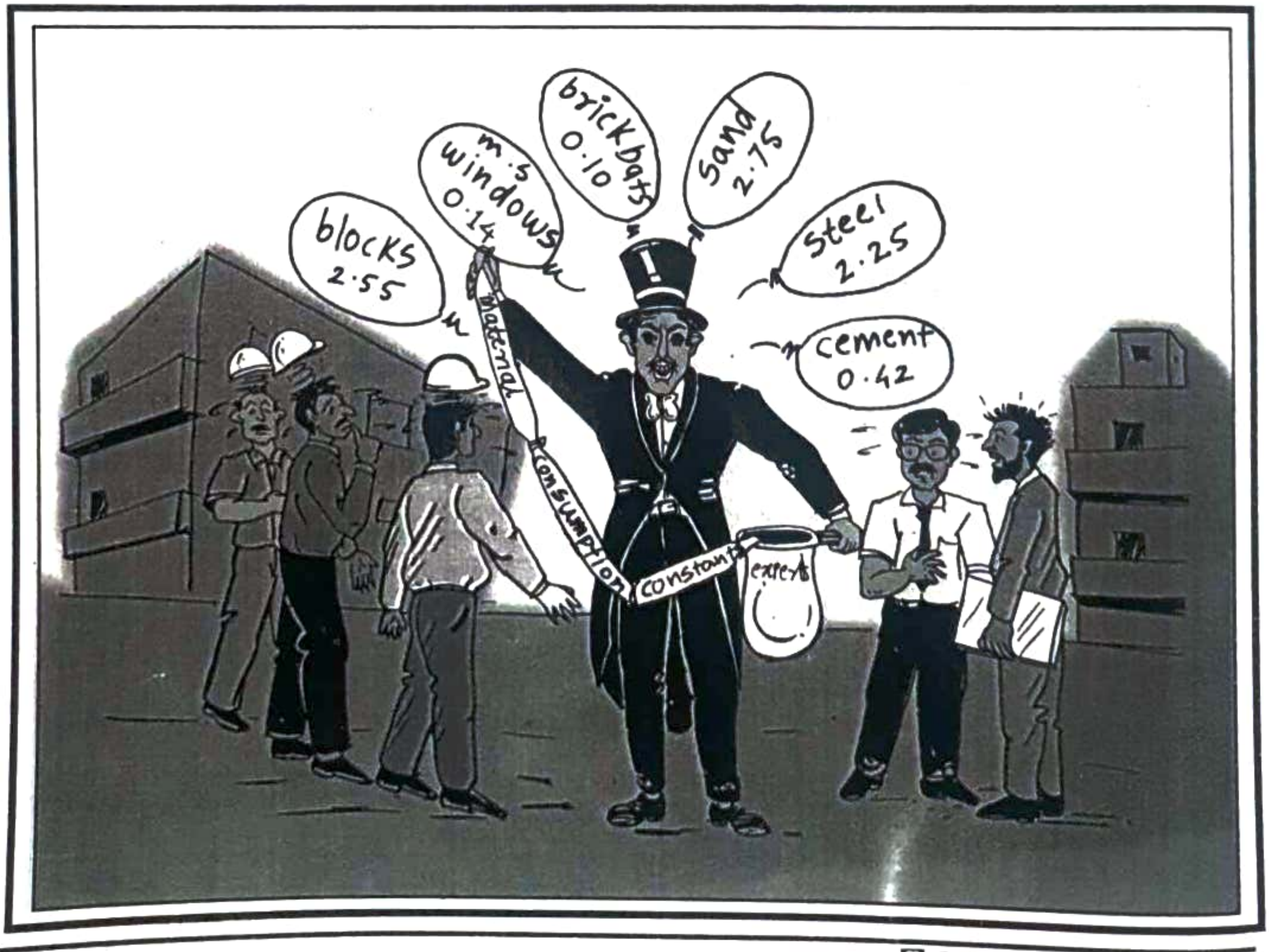
(Refer Table No. 14.4)

SPECIMEN BUILT-UP AREA CALCULATION

TABLE NO. 14.4

FLOOR DETAILS	AREA	CONSIDERED AREA IN %	TOTAL AREA
Parking floor	84 Sq.m.	50%	42 Sq.m.
Ground floor	93 Sq.m.	100%	93 Sq.m.
First floor	207 Sq.m.	100%	207 Sq.m.
Second floor	207 Sq.m.	100%	207 Sq.m.
Third floor	110 Sq.m.	100%	110 Sq.m.
Overhead water tank	7 Sq.m.	50%	3.5 Sq.m.
	TOTAL		662.5 Sq.m. Say 662 Sq.m. = 7128.5 S.ft. Say 7130 S.ft.

'QUICK METHOD ESTIMATION' OF PROJECT AT A GLANCE [ON BUILT-UP AREA BASIS]
(Refer Table No. 14.5 to 14.9 and Figure No. 14.2 to 14.6)



1) DIRECT AND INDIRECT COST BREAK-UP OF BUILDING

TABLE NO. 14.5 (Considering completed project as on 1st February 2001)

Sr. No.	DESCRIPTION	RATE/Sq.m. Approximate rounded (In Rs.)	RATE/SFT Approximate rounded (In Rs.)	COST PERCENTAGE
[A]	**LAND COST**	At actuals	At actuals	———
[B] 01	**DIRECT COST** Material Expenses (Refer Table No. 14.6 for details	3013.00	280.00	63.60%
02	Labour Expenses (Refer Table No. 14.8 for details)	1162.00	108.00	24.50%
[C] 01	**INDIRECT COST** Sanction Expenses	86.00	8.00	1.80%
02	Plot Development	22.00	2.00	0.50%
03	Site Development	108.00	10.00	2.30%
04	Consultancy Expenses	75.00	7.00	1.60%
05	Administrative Expenses	86.00	8.00	1.80%
06	Supervision Expenses	65.00	6.00	1.40%
07	Site Running Expenses	54.00	5.00	1.10%
08	Add 1.50% Contingencies On Total (B)	65.00	6.00	1.40%
	TOTAL COST (B+C)	**4736.00**	**440.00**	**100.00%**

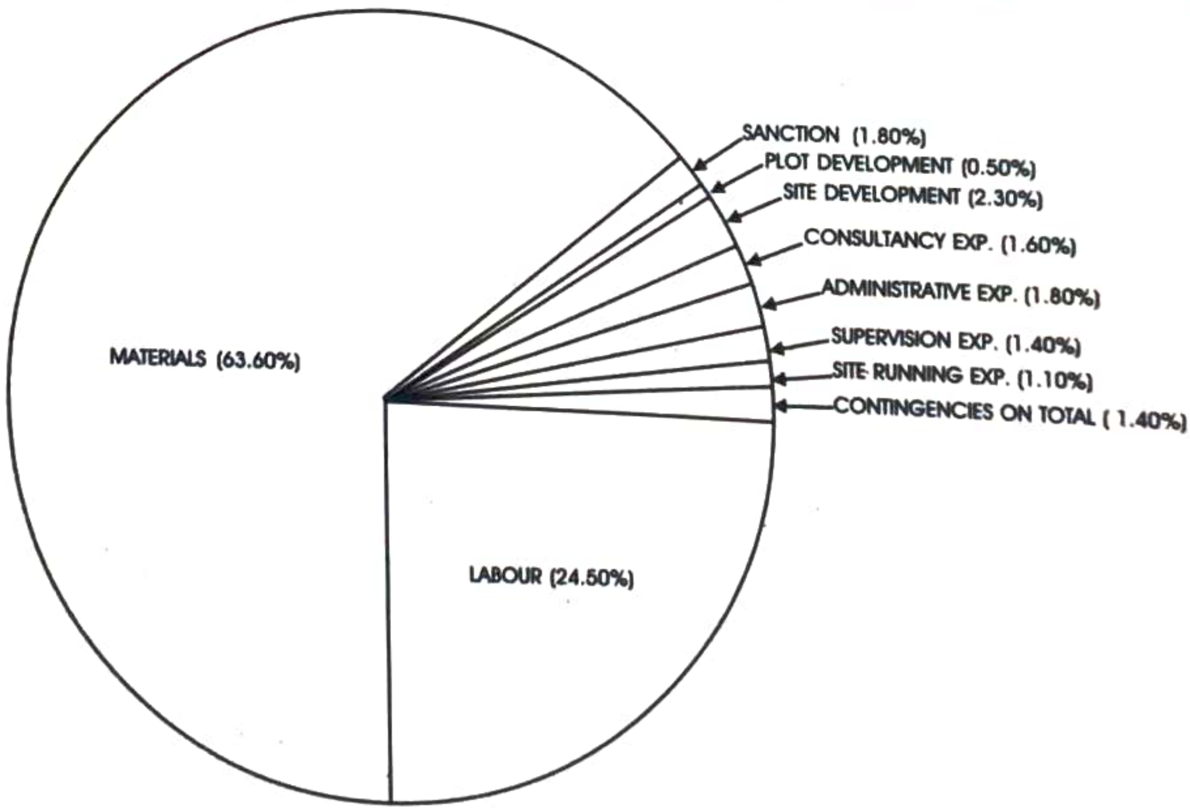


Fig. No. 14.2 : Direct and indirect cost break-up in precentage

2) MATERIAL COST BREAK-UP

TABLE NO. 14.6 (Considering completed project as on 1st February 2001)

Sr. No.	DESCRIPTION	COST IN RUPEES PER		PERCENTAGE
		SQM	SFT	
01	Cement	560.00	52.00	18.60%
02	Steel	377.00	35.00	12.50%
03	Sand	323.00	30.00	10.70%
04	Aggregates (Metal)	140.00	13.00	4.60%
05	Bricks	280.00	26.00	9.30%
06	Wood (Door frames and shutters)	269.00	25.00	8.90%
07	Windows and Glazing	129.00	12.00	4.30%
08	Fooring/Dado/Kitchen otta	301.00	28.00	10.00%
09	Plumbing/Sanitary/Drainage	269.00	25.00	8.90%
10	Electrical Material	129.00	12.00	4.30%
11	Painting Mterial	75.00	7.00	2.50%
12	Other building materials [Rubble, murum, bricks bats Lime, Sanla etc.]	108.00	10.00	3.60%
13	Miscellaneous	54.00	5.00	1.80%
	TOTAL	3014.00	280.00	100.00%

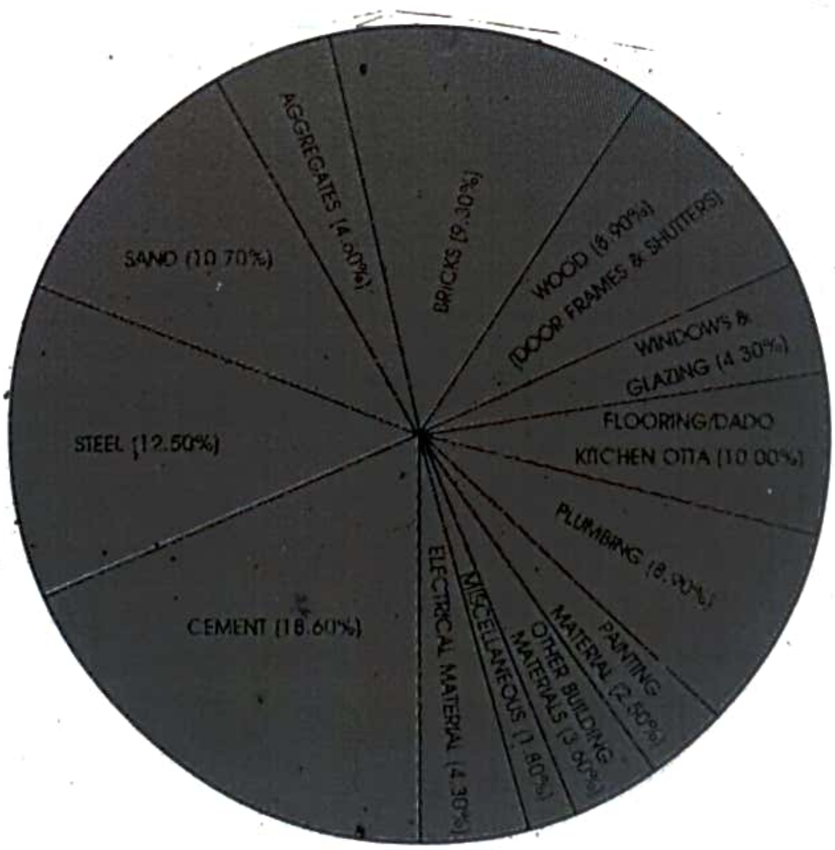


Fig. No. 14.3 : Material cost break-up

3) COST BREAK-UP OF PLUMBING MATERIAL

TABLE NO. 14.7 (Considering completed project as on 1st February 2001)

Sr. No.	DESCRIPTION	COST PER SQM. (IN RUPEES)	COST PER SFT. (IN RUPEES)	PERCENTAGE COST
01	G.I. pipes and fittings	60.30	5.60	22.40%
02	Sanitary ware	47.30	4.40	17.60%
03	P.V.C. pipes and fittings	33.40	3.10	12.40%
04	External water supply	51.60	4.80	19.20%
05	External drainage work	33.40	3.10	12.40%
06	C.P. fittings	43.00	4.00	16.00%
	TOTAL	**269.00**	**25.00**	**100.00%**

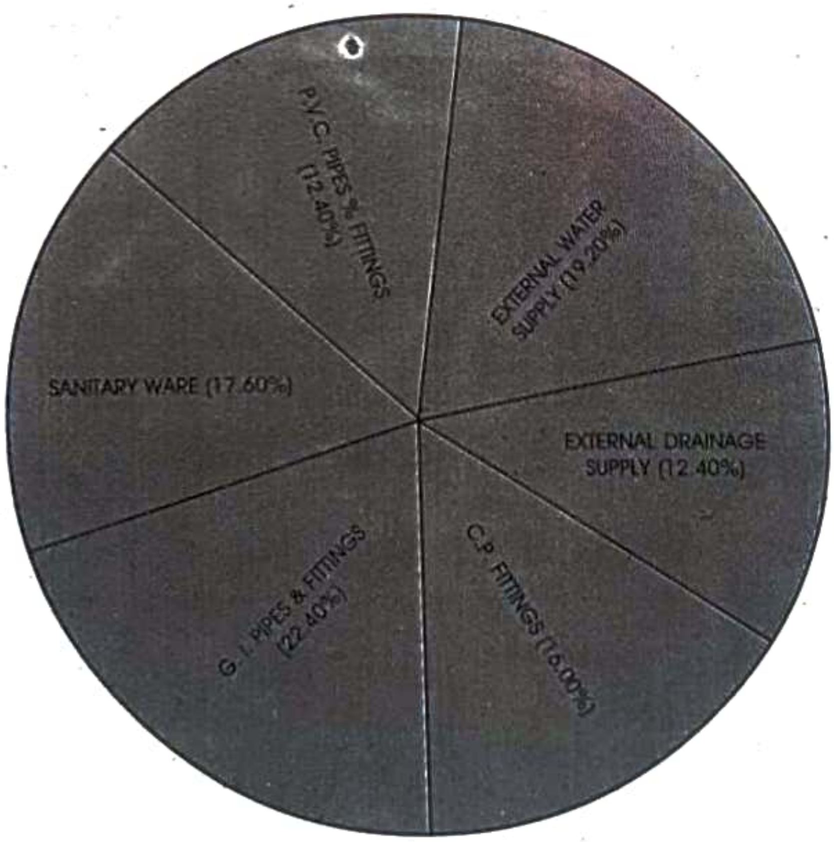


Fig. No. 14.2 : Cost break-up of plumbing mateial

4) LABOUR COST BREAK-UP FOR VARIOUS ITEMS

TABLE NO. 14.8 (Considering completed project as on 1st February 2001)

Sr. No.	DESCRIPTION	RATE Per SQM. (In Rs.)	RATE Per SFT. (In Rs.)	PERCENTAGE
01	R.C.C. work	430.00	40.00	37.00%
02	Masonry and plaster work	280.00	26.00	24.10%
03	Plumbing work	54.00	5.00	4.60%
04	Water proofing work	43.00	4.00	3.70%
05	Carpentry work	32.00	3.00	2.80%
06	Tile fixing work	129.00	12.00	11.10%
07	Electrical work	65.00	6.00	5.60%
08	C.A.T.V. antenna points	22.00	2.00	1.90%
09	Painting work	54.00	5.00	4.60%
10	Departmental labour	54.00	5.00	4.60%
	TOTAL	1163.00	180.00	100.00%

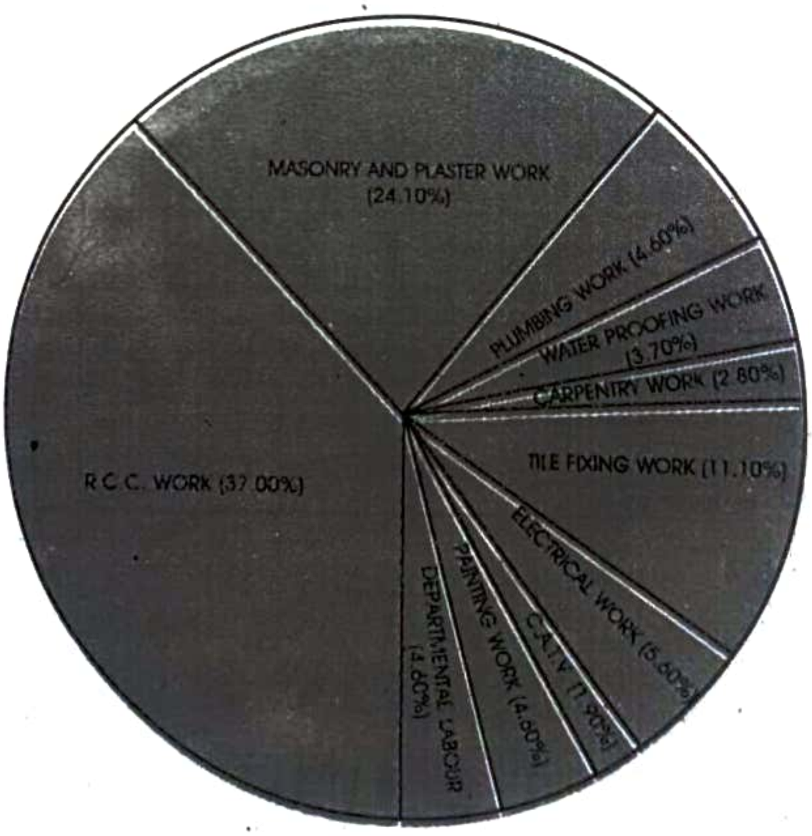


Fig. No. : 14.5 : Labour cost beak-up for various items

5) DIRECT COST BREAK-UP OF BUILDING FOR VARIOUS ITEMS.

(Material + Labour + development)

TABLE NO. 14.9 (Considering completed project as on 1st February 2001)

SR. NO.	DESCRIPTION	RS./ SQM.	RS./ SFT.	PERCNTAGE
01	Plinth	366.00	34.00	8.50%
02	R.C.C. In super structure	1098.00	102.00	25.50%
03	Masonry work with Door frames	560.00	52.00	13.00%
04	Plastering	516.00	48.00	12.00%
05	Water proofing	75.00	7.00	1.80%
06	Flooring and tiling	452.00	42.00	10.50%
07	Door shutter / Windows	377.00	35.00	8.80%
08	Plumbing/sanitation	334.00	31.00	7.80%
09	Electrical work	215.00	20.00	5.00%
10	Painting	129.00	12.00	3.00%
11	Deptt. Labour	54.00	5.00	1.30%
12	Development	129.00	12.00	3.00%
	TOTAL	**4305.00**	**400.00**	**100.00%**

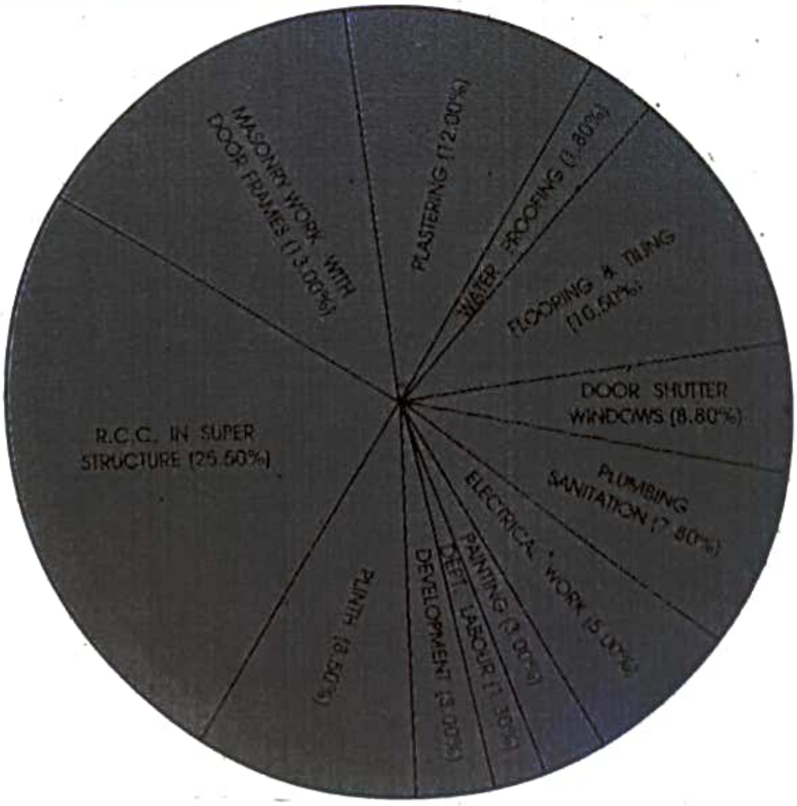


Fig. No. : 14.5 : Direct cost break-up for various items

14.6 DETAILED ESTIMATION

(1) INTRODUCTION

This estimation is also known as the estimate on item rate basis. It is an accurate estimation consisting of the working quantities and rates of each item correctly from the drawing. Quantities of each item are calculated and then abstracting is done. The detailed estimate is prepared in two stages.

a) Taking out measurement and calculation of quantities

The details of measurements of each item of work are taken out from the plans and drawings. Quantities under each item are computed or calculated in a tabular form called the measurement sheet.

b) Abstract of estimated cost

The cost of each item of work is calculated from the quantities and unit rate of item. The rate of finished item of work is analyzed from the present market rates.

Usually 3% estimated cost is added for contingencies such as miscellaneous items that do not come under classified heads of items of the work. The grand total thus added gives the estimated cost of work.

(2) DATA REQUIRED FOR DETAILED ESTIMATION

- Generally, the following data is required for detailed estimation.
  - Detailed working drawings including plan elevation section, site plan, index plan etc.
  - Strata details
  - R.C.C. drawings for foundation, beams, slabs lintels etc.
  - Detailed specifications.
  - General specifications.
  - Analysis of rates, if rates are not as per schedule
  - Location and working conditions at work site

- **REPORTS**

The estimate is usually accompanied by a report detailing all the information about the project in brief. The report should give a clear picture of the whole project/work. It should consist of,

- Brief history with reference to the proposal.
- Location of site.
- Survey reports.
- Miscellaneous items like labour amenities temporary accommodation for staff etc. (for big projects).
- Nature of soil and topography of land.
- The total cost and source of finance.
- Completion period.

- Detailed estimate is mainly prepared for
  - Technical sanction of the competent authority
  - For preparing contract document
  - Execution of the work
  - Calculation of quantities
  - Planning of budget
  - Programming and planning of work
  - Bar chart an other material schedules
  - Arranging for the labour

(3) POINTS TO BE CONSIDERED FOR DETAILED ESTIMATION

- The entire work should be divided into different classes of items. Similar items are grouped under sub-heads of work.
- Norms for calculation of quantities may vary from site to site and according to the rules and regulations of local authorities.
- As far as possible, standard specified norms should be followed.
- Mode of measurements and units may be as per the local practice.

FOLLOWING FORMULAE, ASSUMPTIONS, NORMS SHOULD BE FOLLOWED FOR CALCULATION OF QUANTITIES

- **EXCAVATION**

Consider 15cm extra on either side than the actual size of P.C.C.
Quantity of excavated loose material will be 1.5 times more than the quantity of excavation.

- **PLINTH FILLING**

Quantity of filling material should be considered as 1.5 times more than required volume in the plinth.

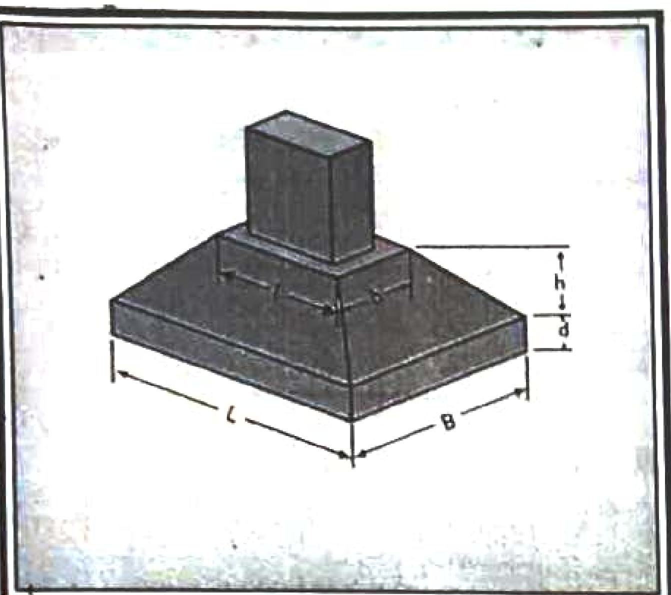


Fig. No. 14.7 : Trapozoidal footing

CONCRETE QUANTITY FOR FOOTING

(Refer Figure no. 14.7)

$V = L \times B \times D + h/3\ (A_1 + A_2 + \sqrt{A_1 A_2})$

- V = Volume of footing
- L = Length
- B = Breadth
- d = Depth of rectangular portion (= column length + 15cm)
- b = breadth of top portion of footing (= column breadth + 15cm)
- h = Height of trapezoidal portion
- A_1 = Area of bottom portion of footing (L x B)
- A_2 = Area of top portion of footing (l x b)

CONCRETE QUANTITY OF COLUMN

L x B x h where

- L = Length of column
- B = Breadth of column
- h = Height of column floor slab to floor slab

CONCRETE QUANTITY OF BEAM/SLAB

- Length of the beam should be considered as clear span of beam [between two columns / supports].
- Depth of beam should be considered up to the top of the slab.
- Slab area should be measured for clear span of slab.

For lintel consider bearing on both the ends and minimum 23cm/as specified.

STEEL QUANTITY

- Weight of bar in kg/meter should be calculated as **(d^2 + 162)** where 'd' is the diameter of bar in mm.
- 'L' for column main steel in footing should be considered as minimum 30cm/as specified.
- For bent up bar add 0.42d in straight length of bar where 'd' is the clear depth of beam/slab.
- For cantilever, beam anchorage length for main steel should be 69D or length of cantilever whichever is greater.

Where
D = diameter of bar in mm.

- $\text{No. of Stirrups} = \dfrac{\text{Clear Span}}{\text{Spanning}} + 1$
- Lap length in column should be 45D
  Where
  D = diameter of bar in mm
- Lap length in beam/slab should be 69D
  Where
  D = diameter of bar in mm

MASONRY AND PLASTER

- For opening in masonry, deduct full clear openings.
- For plaster, deduct half area of opening from internal as well as from the external side and do not add any jams of doors and windows.

PAINTING

For painting consider following multiplying factors for,

- M.S. windows 1.5 times
- Panelled door 2.5 times
- Flush door 2.25 times
- Simple grill 0.5 times
- Rolling shutter 3 times
- Corrugated sheet 3 times
- Collapsible gate 3 times

14.7 STANDARD MATERIAL CONSUMPTION FOR VARIOUS ITEMS

TABLE NO. 14.10

(A)

SR. NO.	ITEMS IN PLINTH	UNIT PER	LOOSE SOIL	LOOSE MURUM	RUBBLE
1	Excavation in black	100 cft	150 cft	——	——
	cotton soil	1 cum	1.5 cum	——	——
2	Compacted murum filling	100 cft	——	150 cft	——
		1 cum	——	1.50 cum	——
3	Rubble soiling 23 cm (9") thick	100 sft	——	——	90 cft
		1 sqm	——	——	0.26 cum

(B)

SR. NO.	CONCRETE ITEMS	UNIT PER	CEMENT	SAND	METAL 25 mm (1")	METAL 20 mm (3/4")	METAL 12 mm (1/2")
1	P.C.C. (1 : 4 : 8)	100 cft	9 bags	60 cft	65 cft	20 cft	——
		1 cum	3.18 bags	0.60 cum	0.65 cum	0.20 cum	——
2	P.C.C. (1 : 3 : 6)	100 cft	12 bags	60 cft	65 cft	20 cft	——
		1 cum	4.24 bags	0.60 cum	0.65 cum	0.20 cum	——
3	R.C.C. (1 : 2 : 4)	100 cft	17 bags	60 cft	——	65 cft	20 cft
		1 cum	6.00 bags	0.60 cum	——	0.65 cum	0.20 cum
4	R.C.C. (1 : 1½ : 3)	100 cft	22 bags	60 cft	——	65 cft	20 cft
		1 cum	7.77 bags	0.60 cum	——	0.65 cum	0.20 cum

(C)

SR. NO.	CONCRETE BLOCKS	UNIT PER	CEMENT	SAND	SHINGLE	METAL 12 mm (1/2")	STONE GRIT	STONE DUST
1	Solid cement concrete block 30 x 15 x 20 cm (12" x 6" x 8")	35 Nos.	1 bag	3.75 cft	2.50 cft	7.00 cft	5.50 cft	2.50 cft
		35 Nos.	1 bag	0.11 cum	0.07 cum	0.20 cum	0.16 cum	0.07 cum
2	Solid cement concrete block 30 x 10 x 20 (12" x 4" x 8")	50 Nos.	1 bag	3.75 cft	2.50 cft	7.00 cft	5.50 cft	2.50 cft
		50 Nos.	1bag	0.11 cum	0.07 cum	0.20 cum	0.16 cum	0.07 cum

(D)									
SR. NO.	MASONRY ITEMS (excluding R.C.C. band	UNIT PER	CEMENT	SAND	RUBBLE	BRICKS 23X10X7 cm	BRICKS 23X15X9 cm	BLOCKS 30X15X20 cm	BLOCKS 30X10X20 cm
1	U.C.R. masonry	100 cft	4.50 bags	40 cft	125 cft	—	—	—	—
	in C.M. 1: 6	1 cum	1.60 bags	0.40 cum	1.25 cum	—	—	—	—
2	23 cm (9") B.B.M.	100 sft	2.50 bags	20 cft	—	1000 Nos.	—	—	—
	in C.M. 1 : 6	1sqm	0.27 bag	0.06 cum		110 Nos.	—	—	—
3	15 cm (6") B.B.M.	100 sft	1.50 bags	15 cft	—	—	400 Nos.	—	—
	in C.M. 1 : 5	1 sqm	0.16 bag	0.05 cum	—	—	45 Nos.	—	—
4	10 cm (4") B.B.M.	100 sft	1.00 bag	8 cft	—	475 Nos.	—	—	—
	in C.M. 1 : 4	1 sqm	0.11 bag	0.25 cum	—	50 Nos.	—	—	—
5	15 cm (6") concrete block masonry	100 sft	1.00 bag	11 cft	—	—	—	145 Nos.	—
	in C.M. 1 : 6	1sqm	0.11 bag	0.03 cum	—	—	—	16 Nos.	—
6	10 cm (4") concrete block masonry	100 sft	1.00 bag	8 cft	—	—	—	—	145 Nos.
	in C.M. 1 : 4	1 sqm	0.11 bag	0.25 cum	—	—	—	—	16 Nos.

(E)					
SR. NO.	PLASTERING ITEMS	UNIT PER	CEMENT	SAND	SANLA (30 Kg bag)
1	Sanla finish plaster 12.5 mm (1/2") thick a) In C.M. 1 : 4	100 sft	1.50 bags	11.0 cft	0.60 bag
		1 sqm	0.16 bag	0.03 cum	0.07 bag
	b) In C.M. 1 : 5	100 sft	1.20 bags	11.0 cft	0.60 bag
		1 sqm	0.13 bag	0.03 cum	0.07 bag
	c) In C.M. 1 : 6	100 sft	1.00 bag	11.0 cft	0.60 bag
		1 sqm	0.11 bag	0.03 cum	0.07 bag
2	Sanla finish plaster 20 mm (3/4") thick a) In C.M. 1 : 4	100 sft	2.4 bags	18 cft	0.60 bag
		1 sqm	0.26 bag	0.06 cum	0.07 bag
	b) In C.M. 1 : 5	100 sft	1.90 bags	18 cft	0.60 bag
		1 sqm	0.21 bag	0.06 cum	0.07 bag
	c) In C.M. 1 : 6	100 sft	1.60 bags	18 cft	0.60 bag
		1 sqm	0.17 bag	0.06 cum	0.07 bag
3	Sand faced plaster Single coat 15 to 20 mm thick a) In C.M. 1 : 4	100 sft	2.25 bags	17.0 cft	—
		1 sqm	0.24 bag	0.05 cum	—
	b) In C.M. 1 : 5	100 sft	1.8 bags	17.0 cft	—
		1 sqm	0.19 bags	0.05 cum	—
	c) In C.M. 1 : 6	100 sft	1.5 bags	17.0 cft	—
		1 sqm	0.16 bag	0.05 cum	—
4	Sand faced plaster Double coat 20 to 25 mm thick a) In C.M. 1 : 4	100 sft	3.75 bags	28.0 cft	—
		1 sqm	0.41 bag	0.09 cum	—
	b) In C.M. 1 : 5	100 sft	3.0 bags	28.0 cft	—
		1 sqm	0.32 bag	0.09 cum	—
	c) In C.M 1 : 6	100 sft	2.5 bags	28.0 cft	—
		1 sqm	0.27 bag	0.09 cum	—

(F)							
SR. NO.	FLOORING & TILING	UNIT PER	CEMENT	SAND	LIME (30 kg/bag)	TILES	WHITE CEMENT
1	Mosaic tile flooring (25 cm x 25 cm)	100 sft	1.0 bag	16 cft	2.0 bags	150 Nos.	5kg
		1 sqm	0.11 bag	0.05 cum	0.2 bag	16 Nos.	0.55 kg
2	Mosaic tile skirting (25 cm x 12.5 cm)	100 rft	2.0 bags	2 cft	——	125 Nos.	2 kg
		10 rm	0.65 bag	0.01 cum	——	14 Nos.	0.65 kg
3	Stone flooring (marble, kotah, tandoor etc. (45 cm x 60 cm uncut size) (after cutting 40 cm x 55 cm)	100 sft	2.0 bags	16 cft	——	43 Nos.	2 kg
		1 sqm	0.22 bag	0.05 cum	——	5 Nos.	0.221 kg
4	Rough shahabad stone flooring (45 cm x 60 cm uncut size) (after cutting 40 cm x 55 cm)	100 sft	0.75 bag	18 cft	2.0 bags	40 Nos.	——
		1 sqm	0.08 bag	0.06 cum	0.2 bag	4 Nos.	——
5	Chequered tile flooring (25 cm x 25 cm)	100 sft	1.25 bags	16 cft	2.0 bags	150 Nos.	——
		1 sqm	0.13 bag	0.05 cum	0.2 bag	16 Nos.	——
6	Crazy (Tukda) mosaic flooring	100 sft	2.50 bags	16 cft	2.0 bags	required tukda	——
		1 sqm	0.27 bag	0.05 cum	0.2 bag	required tukda	——
7	I.P.S. flooring	100 sft	3.5 bags	16 cft	——	——	——
		1 sqm	0.38 bag	0.05 cum	——	——	——
8	Glazed tile dado (15 cm x 15 cm)	100 sft	3.5 bags	——	——	420 Nos.	2 kg
		1 sqm	0.38 bag	——	——	45 Nos.	0.22 kg

(F)								
SR. NO.	WATER PROOFING	UNIT PER	CEMENT	SAND	BRICKBAT	LIME (30 kg/bag)	KAVADI (in kg)	WHITE CEMENT
1	Brickbat coba W/P in W.C., bath, toilet etc.	100 sft	6.0 bags	30 cft	30 cft	——	——	——
		1 sqm	0.65 bag	0.09 cum	0.09 cum	——	——	——
2	Brickbat coba W/P to terrace	100sft	4.0 bags	30 cft	30 cft	——	——	——
		1 sqm	0.43 bag	0.09 cum	0.09 cum	——	——	——
3	Glazed china mosaic (Kavadi) type W/P to terrace (including brick - bat coba)	100 sft	5.0 bags	45 cft	30 cft	1.20 bags	125 kg	25 kg
		1 sqm	0.50 bag	0.14 cum	0.09 cum	0.13 bag	13.5 kg	2.7 kg

14.8 *MODE OF MEASUREMENTS*

Mode of measurements is the method of taking measurements at a site. Mode of measurement of each item varies from place to place as per practical convenience in that particular area. If local labourers are more accustomed with a particular unit then that should be used as the mode of measurement. Mode of measurement given in the table no. 14.11 may not tally with standard specified units.

2) CALCULATIONS OF SHUTTERING MATERIAL REQUIREMENT BY QUICK METHOD ESTIMATION FOR 232 sqm (2500 sft) OF SLAB AREA

TABLE NO. 14.13

SR.	MATERIAL REQUIRED	UNIT	MULTIPLYING CONSTANT BY THUMB RULE	APPROXIMATE QUANTITY OF MATERIAL REQUIRED
1	Plywood 12 m thick of size 2.44 m x 1.22 m (8'-0" x 4'-0")	2500 sft	0.02	50 Nos. Sheets
		232 sqm	0.22	50 Nos. Sheets
2.	Silver oak battern of size 75 mm x 40 mm (3" x $1\frac{1}{2}$")	2500 sft	50 sheets of plywood x 65	3250 rft
		232 sqm	50 sheets of plywood x 19.82	990 rm
3	Beam bottom planks 40 mm thick ($1\frac{1}{2}$")	2500 sft	2500 sft x 0.24	600 rft
		232 sqm	232 sqm x 0.79	183 rm
4	Wooden props a) For beams	2500 sft	600 rft beam bottom x 0.5	300 Nos.
		232 sqm	183 rm beam bottom x 1.65	300 Nos.
	b) For slab	2500 sft	2500 sft x 0.16	400 Nos.
		232 sqm	232 sqm x 1.72	400 Nos.
5	Wooden batten for supporting slab plates 75 mm x 75 mm (3" x 3")	2500 sft	2500 x 0.4	1000 rft
		232sqm	232 x 1.31	305 rm
6	Steel plates of size 60 cm x 90 cm (2'-0" x 3'-0")	2500 sft	2500 x 0.16	400 Nos.
		232 sqm	232 x 1.72	400 Nos.
7	Wire nails	250 sft	2500 x 0.020	50 kg
		232 sqm	232 x 0.22	50 kg
8	Shuttering oil	2500 sft	2500 x 0.006	15 litres
		232 sqm	232 x 0.065	15 litres
9	M.S. clamps (Shikanja)	2500 sft	2500 x 0.08	200 Nos.
		232 sqm	232 x 0.86	200 Nos.

3) CONSUMPTION OF STEEL IN KG PER SQM OR SFT OF WORK

TABLE NO 14.14 (Considering specimen building, Refer Figure No. 14.1)

SR. NO.	ITEMS	CONSUMPTION OF STEEL on built-up area	
01	Footing	0.55 kg/sqm	0.05kg/sft
02	Columns up to plinth	2.15 kg/sqm	0.20 kg/sft
03	Tie beam	1.15 kg/sqm	0.10 kg/sft
04	Column (super structure)	4.80 kg/sqm	0.45 kg/sft
05	Slab	7.0 kg/sqm	0.65 kg/sft
06	Beam	7.5 kg/sqm	070 kg/sft
07	Chajja/loft/lintel	0.55 kg/sqm	0.05 kg/sft
08	Over head water tank	0.55 kg/sqm	0.05 kg/sft
	TOTAL	**24.25 kg/sqm**	**2.25 kg/sft**

4) CONSTANTS FOR ESTIMATING THE APPROXIMATE QUANTITIES OF VARIOUS ITEMS PER SQ. M. / SQ. FT. OF BUILT-UP AREA.

TABLE NO. 14.15 (Considering specimen building Refer Fig. No. 14.1)

SR. NO.	ITEM	APPROX. QTY. PER SQ. M.	APPROX. QTY. PER SQ. FT.	SR. NO.	ITEM	APPROX. QTY. PER SQ. M.	APPROX. QTY. PER SQ. FT.
01	Excavation	0.31 cum	1.00 cft	09	Flooring		
02	R.C.C. below plinth	0.046 cum	0.15 cft		a) MM tile	0.75 sqm	0.75 sft
03	R.C.C. above plinth	0.290 cum	0.95 cft		b) skirting	0.82 rm	0.25 rft
04	Internal Masonry	0.7 sqm	0.7 sft		c) W.C./Bath	0.05 sqm	0.05 sft
05	External Masonry	0.9 sqm	0.9 sft		d) Dado	0.20 sqm	0.20 sft
06	Internal Plaster/Paint	3.2 sqm	3.2 sft	10	Waterproofing	0.30 sqm	0.30 sft
07	External Plaster/Paint	2.5 sqm	2.5 sft	11	Door Frames	0.17 sqm	0.17 sft
08	Oil Paint	0.62 sqm	0.62 sft	12	Windows	0.14 sqm	0.14 sft

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